

- 9** **(a)** When electrons striking the metal target interact with the crystal lattice, forces experienced by the electrons cause them to be accelerated, decelerated or deflected. When this occurs, their kinetic energies are lost B1
through the emission photons of a range of energies which can lie in the X-ray region.

Since the magnitude of the 'deceleration' experienced by the incident electrons is different for all and is not discrete, the wavelengths of the emitted photons have a continuous distribution and produces a continuous spectrum of electromagnetic radiation. B1

- (b)** Accelerated incident electron 'collides' with an electron from the innermost shell kicking it out of the K-shell. The atom is excited due to the vacancy in the K-shell. B1

An electron from the M shell transits to the vacancy in the K-shell, emitting a photon. B1

- (c)** There is a larger occurrence of electrons transiting from L-shell to K-shell as compared to electrons transiting from M-shell to K-shell. B1